

MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 1

Due date: 12 September (Friday), 2025 10:00pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Let U be an upper triangular matrix. Suppose all the diagonal entries of U are zero. Show that there exists an integer N such that for all $k > N$, $U^k = 0$.
2. Let K be the $n \times n$ matrix whose entries are 1 on the anti-diagonal entries and 0 elsewhere. The examples are: for $n = 2$,

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

for $n = 3$,

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

for $n = 4$,

$$\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Show that if U is a lower triangular $n \times n$ matrix, then KUK is an upper triangular matrix.

3. A matrix P is said to be a permutation matrix if for each row and column of P , there is precisely one non-zero entry in that row or column and such entry is one. Examples of permutation matrices include the identity matrix and the matrix K in Question 2. Other examples include

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Let A be an $n \times n$ matrix and let P be an $n \times n$ permutation matrix. Show that the number non-zero entries in A is the same as the number of non-zero entries in PAP .

MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 2

Due date: 22 September (Monday), 2025 10:00pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Let A be an invertible 5×5 matrix and let B be an invertible 3×3 matrix. Let C be a 5×3 matrix. Consider the following block matrix:

$$D = \begin{pmatrix} A & C \\ 0_{3 \times 5} & B \end{pmatrix}.$$

Determine if D is invertible. If D is invertible, find D^{-1} in terms of A^{-1} , B^{-1} , A , B , C . If D is not always invertible, give an example.

2. Let v be an $n \times 1$ matrix. Suppose $n \geq 2$. Use properties of determinants to prove that $\det(vv^T) = 0$.

**MATH 2101 LINEAR ALGEBRA I, FALL 2025 – OUTLINE OF
SOLUTIONS FOR ASSIGNMENT 2**

Due date: 22 September (Monday), 2025, 10:00 pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
 - All solutions have to be turned in HKU moodle in the format of *PDF file*.
 - Please include your *Name, UID, Faculty, Major (if declared)* in your solution.
1. Let A be an invertible 5×5 matrix and let B be an invertible 3×3 matrix. Let C be a 5×3 matrix. Consider the following block matrix:

$$D = \begin{pmatrix} A & C \\ 0_{3 \times 5} & B \end{pmatrix}.$$

Determine if D is invertible. If D is invertible, find D^{-1} in terms of A^{-1} , B^{-1} , A , B , C . If D is not always invertible, give an example.

Sol: We claim that the following 8×8 matrix

$$E = \begin{pmatrix} A^{-1} & -A^{-1}CB^{-1} \\ 0_{3 \times 5} & B^{-1} \end{pmatrix}$$

is the inverse of D . In particular, D is invertible. Indeed, we will verify that $ED = DE = I_8$. Using the multiplication rule for block matrices, and the fact that A and B are invertible, one has

$$\begin{aligned} ED &= \begin{pmatrix} A^{-1} & -A^{-1}CB^{-1} \\ 0_{3 \times 5} & B^{-1} \end{pmatrix} \begin{pmatrix} A & C \\ 0_{3 \times 5} & B \end{pmatrix} \\ &= \begin{pmatrix} A^{-1}A & A^{-1}C - A^{-1}CB^{-1}B \\ 0_{3 \times 5} & B^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} I_{5 \times 5} & 0_{5 \times 3} \\ 0_{3 \times 5} & I_{3 \times 3} \end{pmatrix} = I_8. \end{aligned}$$

Similarly, one has

$$\begin{aligned} DE &= \begin{pmatrix} A & C \\ 0_{3 \times 5} & B \end{pmatrix} \begin{pmatrix} A^{-1} & -A^{-1}CB^{-1} \\ 0_{3 \times 5} & B^{-1} \end{pmatrix} \\ &= \begin{pmatrix} AA^{-1} & -AA^{-1}CB^{-1} + CB^{-1} \\ 0_{3 \times 5} & BB^{-1} \end{pmatrix} \\ &= \begin{pmatrix} I_{5 \times 5} & 0_{5 \times 3} \\ 0_{3 \times 5} & I_{3 \times 3} \end{pmatrix} = I_8. \end{aligned}$$

Therefore, we conclude that $D^{-1} = E$, which can be expressed using A^{-1} , B^{-1} , A , B and C as above.

Remark. One way to obtain E is to assume it is of the form

$$E = \begin{pmatrix} X & Y \\ Z & W \end{pmatrix},$$

which is of the same block structure as D , i.e., X and A have the same block size, Y and C have the same block size etc. Upon stipulating that $ED = I_8$, we can obtain E in the form above. However, one still has to use the definition of matrix inverse to show that $E = D^{-1}$.

2. Let v be an $n \times 1$ matrix. Suppose $n \geq 2$. Use properties of determinants to prove that $\det(vv^T) = 0$.

Sol: Let $v = (a_1, a_2, \dots, a_n)^T$. Then

$$\begin{aligned} \det(vv^T) &= \det \begin{pmatrix} a_1 v^T \\ a_2 v^T \\ \vdots \\ a_n v^T \end{pmatrix} \\ &= a_1 a_2 \dots a_n \det \begin{pmatrix} v^T \\ v^T \\ \vdots \\ v^T \end{pmatrix} \\ &= 0, \end{aligned}$$

where the first equality follows from matrix multiplication, the second equality follows from the scalar rule in computing determinants and the third equality follows from that there are two identical rows in the matrix.

MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 3

Due date: 2 October (Thursday), 2025 10:00pm.

- To receive *full credits*, the solution has to be *clear* and provide sufficient explanations.
- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Solve the following system linear equations by Jordan-Gaussian eliminations:

$$-x_1 + 2x_2 + x_3 - x_4 + 3x_5 = 7$$

$$x_1 - 4x_2 + 5x_4 - 5x_5 = -7$$

$$-2x_1 + 6x_2 - x_3 - 5x_4 + 7x_5 = 6$$

$$-x_1 - 4x_2 + 4x_3 + 11x_4 - 2x_5 = 11$$

2. Find the reduced row echelon form of the following matrix:

$$\begin{pmatrix} c & 0 & 2 & 2 \\ 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

for all the values of $c \in \mathbb{R}$.

3. Let A be an $n \times m$ matrix and let b be a $n \times 1$ column vector. Suppose all the coefficients of A and B are integers. Determine if the following statements are true or false.

- If the system $Ax = b$ has m leading variables, then any solution for the system $Ax = b$ has rational entries.
- If the system $Ax = b$ has n leading variables, then any solution for the system $Ax = b$ has rational entries.

If the statement is true, explain it. If the statement is false, give a counterexample.

4. Let A be an invertible square 3×3 matrices.

- Show that A is equal to UPL for some lower triangular matrix L , some upper triangular matrix U and permutation matrix P . (See Assignment 1 for the definition of a permutation matrix.)
- Is the expression UPL above unique?

MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 4

Due date: 10 October (Friday), 2025 10:00pm.

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- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. For two vector subspaces W_1, W_2 of a vector space V , define

$$W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\}.$$

Show that $W_1 + W_2$ is a vector subspace of V .

2. Let S be a subset of a vector space V . Show that $\text{span}(S) = S$ if and only if S is a vector subspace of V .
3. Let $\text{Mat}_{5 \times 5}$ be the vector space of 5×5 matrices. Determine if the following sets form a vector subspace of $\text{Mat}_{5 \times 5}$.
 - (a) Let B be a 6×5 -matrix. Let $S_1 = \{A \in \text{Mat}_{5 \times 5} : BA = 0\}$.
 - (b) Let $S_2 = \{A \in \text{Mat}_{5 \times 5} : \det(A) = 0\}$.
 - (c) Let $S_3 = \{A \in \text{Mat}_{5 \times 5} : A^T = A\}$.
 - (d) Let B be a 5×5 -matrix. Let $S_4 = \{A \in \text{Mat}_{5 \times 5} : BA = I_5\}$.

Explain your answers.

4. Determine whether $\begin{pmatrix} 1 \\ 0 \\ 2 \\ 3 \end{pmatrix}$ is in

$$\text{span}\left(\left\{\begin{pmatrix} 1 \\ -2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix}\right\}\right).$$

Explain your answer.

MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 5

Due date: 3rd November (Friday), 2025 10:00pm.

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- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Let $\mathcal{P}_5$ be the set of polynomials with degree less than or equal to 5. It is known that the set

$$S = \{1 + x - x^2, x + x^2, 3x + 2x^4\}.$$

is linearly independent. What is the dimension of $\mathcal{P}_5$? Find a basis for $\mathcal{P}_5$ which contains S and *other monomials*.

2. Let $V = \{(a_1, a_2) : a_1, a_2 \in \mathbb{R}\}$. Define addition on V as usual: for $(a_1, a_2) \in V$ and $(b_1, b_2) \in V$,

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2).$$

Define scalar multiplication on V by: for $c \in \mathbb{R}$ and $(a_1, a_2) \in V$, define

$$c(a_1, a_2) = \begin{cases} (0, 0) & \text{if } c = 0 \\ (\frac{a_1}{c}, ca_2) & \text{if } c \neq 0 \end{cases}$$

Determine whether V is a vector space.

3. Let u, v be vectors in a vector space V . Show that if $u + v, u - v$ are linearly independent, then $u + 2v, u - 2v$ are linearly independent.
4. (a) Let S_1 and S_2 be two sets of linearly independent vectors in a vector space V . Suppose $S_1 \cap S_2 = \emptyset$. Show that $S_1 \cup S_2$ is still linearly independent in V if and only if $\text{span}(S_1) \cap \text{span}(S_2) = \{0\}$.
(b) Let A_1 and A_2 be two $n \times m$ matrices. Let $d_1 = \dim N(A_1)$ and let $d_2 = \dim N(A_2)$. Show that if $d_1 + d_2 > m$, then there is a non-zero *common* solution in $\mathbb{R}^m$ for the two systems $A_1x = 0$ and $A_2x = 0$.
(c) What can one say about the previous question if $d_1 + d_2 \leq m$?

MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 6

Due date: 14th November (Monday), 2025 10:00pm.

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- All solutions have to be turned in HKU moodle in the format of *PDF file*.
- Please include your *Name, UID, Faculty, Major (if declared)* in your solution.

1. Let $\mathcal{P}_5$ be the vector space of polynomials of degree less than or equal to 5. Define $T_1 : \mathcal{P}_5 \rightarrow \mathcal{P}_5$ given by: for any $f \in \mathcal{P}_5$,

$$T_1(f)(x) = f(2x + 1)$$

and

$$T_2(f)(x) = 2f(x) + 1.$$

Determine whether T_1 and T_2 are linear transformations.

2. Let $T : V \rightarrow W$ be an invertible linear transformation. Prove that there exists a basis β for V and a basis γ for W such that $[T]_{\beta}^{\gamma}$ is the identity matrix.
3. Let $T' : V \rightarrow W$ and let $T'' : W \rightarrow X$ be linear transformations. Suppose T'' is injective. Prove that $N(T') = N(T'' \circ T')$.
4. Let M_3 be the vector space of 3×3 matrices. Let

$$P = \begin{pmatrix} 1 & -2 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Define $T : M_3 \rightarrow M_3$ given by:

$$T(A) = PAQ.$$

Find a basis for $R(T)$ and a basis for $N(T)$. Explain your answer. (Hints: Let $T'(A) = AQ$ and $T''(A) = PA$. Consider $T = T'' \circ T'$. What can you tell about T'' ?)

MATH 2101 LINEAR ALGEBRA I, FALL 2025 – ASSIGNMENT 7

Due date: 20th November (Thursday), 2025 10:00pm.

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1. Determine if each of the following statements is true: Let A be a 3×3 matrix.

- (a) If A is diagonalizable, then A^2 is also diagonalizable.
- (b) If A^2 is diagonalizable and A is invertible, then A is also diagonalizable.
- (c) If A^3 has 3 distinct eigenvalues, then A is diagonalizable.

If the statement is true, provide a proof. If the statement is false, provide a counterexample.

2. Determine all the possible values of $x \in \mathbb{R}$ such that the following matrix

$$\begin{pmatrix} 2+x & x \\ -x & -x+2 \end{pmatrix}$$

is diagonalizable.